

Yannik Schuler

Algebraic Geometer & Mathematical Physicist

Email: [yannik.schuler\[at\]math\[dot\]ethz\[dot\]ch](mailto:yannik.schuler[at]math[dot]ethz[dot]ch)

Web: yannikschuler.github.io

Employment

- Jan 2026 – **Walter Benjamin Fellow of the Deutsche Forschungsgesellschaft**
Dec 2027 ETH Zürich, Department of Mathematics
 In Rahul Pandharipande's research group
- Sep 2024 – **Postdoc**
Dec 2025 ETH Zürich, Department of Mathematics
 In Rahul Pandharipande's research group
- Mar 2024 – **Research Associate**
Aug 2024 University of Cambridge, DPMMS
 In Dhruv Ranganathan's research group

Education

- Nov 2020 – **PhD Mathematics**
Jun 2024 University of Sheffield
 Supervisor: Andrea Brini
 Thesis: "Topics in Gromov–Witten theory"
- Apr 2017 – **MSc Physics**
Jun 2020 University of Bonn
 Thesis: "Higher quantum Airy structures in two-dimensional topological gravity"
 under supervision of Albrecht Klemm and Gaëtan Borot
- Apr 2017 – **BSc Mathematics**
Mar 2019 University of Bonn
 Thesis: "Representations of modulated quivers"
 under supervision of Jan Schröer
- Oct 2013 – **BSc in Physics**
Sep 2016 University of Bonn
 Thesis: "Gravitational waves in view of the discovery at LIGO"
 under supervision of Albrecht Klemm

Publications

Papers

- [1] Michel van Garrel, Navid Nabijou, and Yannik Schuler. "Gromov–Witten theory of bicyclic pairs". *Trans. Am. Math. Soc.* 379.5 (2026), pp. 3353–3402. DOI: 10.1090/tran/9581.
- [2] Yannik Schuler. "The log-open correspondence for two-component Looijenga pairs". *Forum Math. Sigma* 13 (2025). Id./No. e102, p. 30. DOI: 10.1017/fms.2025.10042.

- [3] Andrea Brini and Yannik Schuler. “On quasi-tame Looijenga pairs”. *Commun. Num. Theor. Phys.* 17.2 (2023), pp. 313–341. DOI: 10.4310/CNTP.2023.v17.n2.a3.
- [4] Gaëtan Borot, Reinier Kramer, and Yannik Schuler. “Higher Airy structures and topological recursion for singular spectral curves”. *Ann. Inst. Henri Poincaré Comb. Phys. Interact.* (2023). DOI: 10.4171/AIHPD/168.

Pre-Prints

- [5] Yannik Schuler. “Gromov–Witten invariants and membrane indices of fivefolds via the topological vertex”. 2026. arXiv: 2603.24256 [math.AG].
- [6] Andrea Brini and Yannik Schuler. “Refined Gromov–Witten invariants”. 2024. arXiv: 2410.00118 [math.AG].

Talks & Presentations

Jul 2026	Refinements in Enumerative Geometry and Physics, ICTS Bengaluru, “Membranes and Maps”
May 2026	Heidelberg Algebraic Geometry Seminar, “Membranes and Maps”
May 2026	Hilbert in Mirror, SIMIS, Shanghai, “Fivefold perspective on the GW/DT correspondence”
Nov 2025	Physical Mathematics Seminars, Geneva, “Equivariant topological strings on fivefolds”
Oct 2025	Harvard–MIT Algebraic Geometry Seminar, “Stable maps to fivefolds”
Oct 2025	Geometry, Symmetry, and Physics Seminar, Yale University, “Refined Gromov–Witten invariants”
Oct 2025	Informal Mathematical Physics Seminar, Columbia University, “Membranes and Maps”
Oct 2025	Algebraic Geometry and Moduli Seminar, ETH Zürich, “Quintuple Hodge Integrals”
Aug 2025	Moscow–Beijing topology seminar, online, “Refined Gromov–Witten invariants”
Jun 2025	Helvetic Algebraic Geometry Seminar, Les Diablerets, “Curves in 5-folds and conjectures on Hodge integrals”
Jan 2025	Algebra and Quantum Geometry of BPS quivers, Les Diablerets, “Membranes and Maps”
Sep 2024	Algebraic Geometry and Moduli Seminar, ETH Zürich, “Refined mirror symmetry of local $\mathbb{P}^2$ ”
Oct 2024	Matamzee, “Maps, Sheaves and Membranes”
Sep 2024	Duality, Geometry and Physics, Bernoulli center, Lausanne, “Maps, Sheaves and Membranes”
Sep 2024	Geometry and Physics of Mirror Symmetry, Sheffield, “A worldsheet definition of the refined topological string”
May 2024	Fanosearch Seminar, Imperial College, “Log vs Open Gromov–Witten theory”
May 2024	Calf meeting, University of Bath, “Tangents to the nodal cubic”
May 2024	Real and Logarithmic Enumerative Geometry, Oberwolfach, “Gromov–Witten theory and the refined topological string”
Apr 2024	Junior Geometry Seminar, University of Cambridge, “Tangents to the nodal cubic”
Apr 2024	Algebra Seminar, University of Edinburgh, “Gromov–Witten theory and the refined topological string”
Dec 2023	Algebraic Geometry and Moduli Seminar, ETH Zürich, “Gromov–Witten theory and the refined topological string”
Nov 2023	Mathematical Physics Seminar, HU Berlin, “Gromov–Witten theory from the fivefold perspective”
Nov 2023	Algebraic Geometry Seminar, University of Utrecht, “Equivariant Strings and Gromov–Witten theory”
Nov 2023	MAGIC seminar, Imperial College, “Equivariant Curves and Strings”
Oct 2023	Sheffield Geometry and Physics Seminar, University of Sheffield, “Tangents to the Nodal Cubic”
Jul 2023	LMS Summer School 2023, University of Sheffield, “From Counting Curves to Strings and back”

May 2023	Geometry and Mathematical Physics Seminar, University of Birmingham, “Gromov–Witten theory from the fivefold perspective”
Feb 2023	Algebraic Geometry Seminar, University of Cambridge, “The Log-Local-Open Correspondence”
Sep 2022	Conference TRSalento22, Otranto, Italy, Poster on “The Log-Local-Open Correspondence”
Jul 2022	K-theory, algebraic cycles and motivic homotopy theory, INI Cambridge, “The Log-Local-Open Correspondence”
Apr 2022	Mirrors, Moduli and M-theory in the Midlands – LMS Regional Meeting & Workshop, University of Birmingham, “Correspondences in enumerative geometry”

Teaching Experience

Sep 2025 –	Lecturer ETH Zürich
	• Bachelor Seminar “Quiver Representations” (Fall 2025) • Master Seminar “Gromov–Witten theory” (Fall 2026)
Apr 2020 –	Postgraduate Teaching Assistant
Sep 2023	University of Sheffield
	• Demonstrator & Marker for “Core Foundation Mathematics” (Prep Course) • Demonstrator & Marker for “Mathematical modelling” (BSc) • Demonstrator & Marker for “Foundations of Pure Mathematics” (BSc) • Marker for “Advanced Calculus and Linear Algebra” (BSc)
Apr 2015 –	Teaching Assistant
Sep 2020	University of Bonn
	• Demonstrator & Marker for “Mathematics for physicists” (BSc) • Demonstrator & Marker for “Mathematics for computer scientists” (BSc) • Demonstrator & Marker for “Theoretical classical mechanics” (BSc) • Demonstrator & Marker for “Theoretical electrodynamics” (BSc) • Demonstrator & Marker for “Theoretical statistical physics” (BSc) • Head Tutor for “General Relativity” (MSc)

Academic service

Referee	Full reports for <i>Annales Henri Poincaré</i> , <i>Communications in Mathematical Physics</i> , <i>International Mathematics Research Notices</i> , <i>Journal of Algebraic Geometry</i> , <i>Transactions of the American Mathematical Society</i> Quick opinions for <i>Advances in Mathematics</i> , <i>Journal de l'École polytechnique Mathématiques</i>
Reviewer	for zbMATH Open

Scholarships

- Walter Benjamin Fellowship of the Deutsche Forschungsgesellschaft (Jan 2026 – Dec 2027)
- Scholarship of the Bonn-Cologne Grad. School of Physics and Astronomy (Oct 2017 – Sep 2019)
- University of Sheffield Faculty of Science Scholarship (Nov 2020 – May 2024)